

SOLAR ON A SHOESTRING

Passive solar, solar water heating, and PV are making headway in affordable housing design and construction in North Carolina.

by JEFF TILLER

A common and often overlooked problem with affordable housing is high energy use—resulting in high energy bills. These bills are often difficult, if impossible, for many residents on a fixed budget to afford, creating the Catch-22 of an affordable housing unit that residents cannot afford to live in.

In North Carolina, a number of affordable housing coordinators have recognized this problem and are working to turn it around. Appalachian State University (ASU), where I work as the head of construction technology, wanted to support these efforts to dramatically reduce energy bills in affordable housing units. To realize this goal, we wrote a proposal to the state energy office of North Carolina to design highly energy-efficient affordable housing.

The funding was secured, and ASU set to work contacting affordable housing groups to see if they wanted to participate in our program. All of the groups—which included Catawba Valley Habitat for Humanity, the Northwest Regional Housing Authority, Rural Mountain Housing, and other Habitat for Humanity (HFH) chapters in western North Carolina—were enthusiastic and agreed to participate in the project. Our goal was to come up with designs for affordable passive solar homes—some of which were new designs and some of which were redrawn plans for affordable housing organizations that already had homes in the design stage. Our contract



The Baby Cape home has low-e windows throughout the house with reduced glass on east, west, and northern exposures.

also included funding for one Zero-Energy Home, so ASU asked Catawba Valley HFH if they would be interested in building it. They agreed to be the guinea pig.

The project resulted in a plan book of 12 affordable passive-solar home plans, commitments by affordable housing groups to build 30 homes using the plans, and design and construction of a Net Zero-Energy Home in Catawba Valley HFH's new infill development. This home minimizes the need for electricity through a variety of energy efficiency and solar energy measures. The remaining electrical needs will be bal-

anced by electrical power from a PV array located on the roof.

Starting with The Oaks

The project's implementation began with the enthusiastic support of Ned Fowler, executive director of Northwest Regional Housing Authority. The organization's new development, called The Oaks, located near Jefferson, North Carolina, will showcase 18 new homes, almost all of which are being built according to ASU's plans. In addition to their passive-solar features, the homes are designed with energy efficient detailing, solar water heating, and high-efficiency HVAC systems. They also include green building details, such as:

- drought-tolerant landscaping;
- tree planting (minimum of 12 trees per acre);
- milling of cleared trees and vegetation;
- reduction of construction waste using optimum-value engineering for framing details;
- moisture control details;
- low-sone ventilation system;
- radon/soil gas vent system;
- CFL fixtures and Energy Star refrigerator, dishwasher, and clothes washer; and
- low-VOC paints and finishes.

To ensure the homes would be as energy efficient as they were designed to be, the housing authority signed up with Advanced Energy's SystemVision pro-



LAUREL ELM

Landscaping at The Oaks includes such features as a minimum of 12 trees per acre and drought-tolerant planting.

gram (see “Advanced Energy’s SystemVision,” and “Ten Rigorous Standards for SystemVision”). Fowler also chose to pursue certification of the homes in the North Carolina HealthyBuilt Homes program, a green building program developed by the North Carolina Solar Energy Center. Although the homes had a low construction budget, the finished houses qualify for a Bronze rating in the North Carolina HealthyBuilt Homes program.

The builder of the first of these homes, Charles Harris, was a little skeptical at first, but he proved to be an enthusiastic participant in the program. He won the contract for the first two homes and now wants to build the

other ten homes that the housing authority is planning to build directly. Rural Mountain Housing, a Methodist-based affordable housing group, is building the other eight homes. Seven of these will be passive-solar homes; their first home was already under construction when the project began.

Harris has completed the first home and has almost finished the second one. Inspectors from Advanced Energy say that the first home, named the Baby Cape, is one of the most attractive affordable homes they have ever seen. The blower door and duct leakage numbers are impressive for a small home on a builder’s first try at meeting SystemVi-

sion requirements. The Baby Cape has a CFM₅₀ of 956 with a CFM₅₀ /ft² of 0.30, and a CFM₂₅ of duct leakage of 48 (3.5% of the floor area and 6% of the total fan flow rate). The ASU team found that most of the duct leakage was in the mechanical closet, not in the ductwork itself. The housing authority is now beginning its third home in the development. The first two units faced west, so the southern exposure was on the side. The third unit faces due south—looking out directly at nearby Mount Jefferson.

Passive-Solar Design Process

Ideally, a passive-solar home should be as simple as possible. The Baby Cape includes the following key design features:

- reduced glass area on east, west and northern exposures;
- additional glass area on the southern exposure;
- shading for the windows during the summer months—overhangs for direct sunlight and landscaping to minimize ground-reflected and diffuse sunlight;
- thermal storage mass (insulated concrete slab floor);
- activity areas, such as living rooms, dining and breakfast rooms, and kitchens on the south side (although kitchen cabinetry can restrict the area available for south-facing windows);

Advanced Energy’s SystemVision

Advanced Energy Corporation, located in Raleigh, North Carolina, provides innovative solutions to energy and performance problems in industry, electric motors, and buildings.

The SystemVision initiative was launched by Advanced Energy in 2000. Having worked with partners across the country to develop home guarantee programs, Advanced Energy now offers its own Homes Built with SystemVision Guarantee to homeowners of more modest means. In partnership with the North Carolina Community Development

Initiative and the North Carolina Housing Finance Agency, more than 20 non-profit builders in North Carolina, including community development corporations (CDCs) and Habitat for Humanity affiliates, plan to extend this guarantee to their homeowners. With this SystemVision cost and comfort guarantee in hand, the original homeowner is assured that for a period of two years

- the energy used to heat and cool the home will not exceed a specified amount; and

- the temperature in the center of any conditioned room will not vary more than three degrees from the thermostat setting.

If those conditions aren’t met, Advanced Energy picks up the tab for the energy cost overrun and for identifying problems with the original equipment or construction.

For more information, go to www.advancedenergy.org.

Ten Rigorous Standards for SystemVision

1. Air Barrier: (a) There must be a continuous, durable air barrier surrounding the conditioned space. (b) There can be no large openings in the framing between the house and unconditioned spaces.

2. House Airtightness: Must be less than or equal to 0.35 CFM₅₀/ft² of envelope area.

3. Insulation: All insulation must be installed so that the full R-value is achieved in every location. No voids, gaps, compression, or wind intrusion is permitted in wall, floor, or ceiling insulation. The insulation must be touching the air barrier in all locations.

4. Windows: (a) All windows, skylights, door glass, and sliding glass doors must be at least double-paned, low-e with a U-value of 0.40 or less and a SHGC of 0.48 or less. (b) If metal-framed units are used, they must have a thermal break in the frame.

5. Heating, Cooling, and Hot Water Efficiency: (a) Maximum energy use must be 30% less than that of a reference house of the same size and shape built to the standards of the 1993 Model Energy Code. (b) Heat pumps and air conditioners must be sized according to ACCA *Manual J*, and must be within 1/2 ton of those specifications. They must have a SEER rating of at least 12 and a heating season performance factor of at least 7.

6. Air Distribution System: (a) All joints in the air distribution system must be sealed with duct mastic and fiberglass mesh, or an equivalent method that has been approved by Advanced Energy. (b) Total duct leakage must be less than 3% of the floor area. (For example, if the house has 1,000 ft² of conditioned space, there must be no more than 30 CFM of duct leakage when tested). (c) All ducts outside the conditioned space must be insulated to at least R-6. (d) Tested airflow must be within +/- 10% of the designed air flow.

7. Mechanical Equipment Installation: (a) Indoor and outdoor units must be matched according to the

ARI Directory or the manufacturer's listing. (b) Inert gas (nitrogen) must be used during any brazing or soldering of refrigerant lines. (c) Condensation drain must be installed to meet local codes and manufacturer's instructions. (d) There must be proper air flow volume across the indoor coil and/or heat exchanger as specified by the manufacturer's instructions. (e) Correct charge, according to the manufacturer's specifications, must be installed.

8. Ventilation: (a) Each bathroom must have an exhaust fan that exhausts a minimum of 50 CFM directly to the outside. This is the measured flow, as installed, not the rated flow. (b) Each kitchen must have a range hood or exhaust that exhausts a minimum of 100 CFM directly to the outside. This is the measured flow, as installed, not the rated flow. (c) There must be whole-house ventilation equivalent to 10 CFM per bedroom plus 10 CFM continuous. (d) Solid metal, flexible metal, or lined insulated flexible duct can be used. Flexible vinyl ducting cannot be used. (e) All ventilation ducts in unconditioned spaces must be insulated.

9. Pressure Balancing and Return Air Paths: No pressures with a magnitude greater than 3 Pa with reference to the outside may exist in any zone of a house.

10. Combustion Safety: (a) Combustion equipment, including but not limited to furnaces and water heaters, must be located outside the conditioned space, unless it is sealed-combustion or power-vented equipment. (b) In any house with an attached garage or a combustion appliance, one carbon monoxide (CO) detector shall be installed per 1,000 ft² of living space, with a minimum of one detector per floor. (c) Unvented fireplaces are not permitted. (d) Gas logs are not permitted if they have a fan that blows any portion of the combustion gases into the living spaces.

- nighttime spaces, primarily bedrooms, and spaces that have low occupancy, such as closets and bedrooms, on non-southern exposures; and

- no roofs that would shade windows in the winter months, such as porches on the south side.

Finding the right glass for the southern exposure proved to be a bit more complex than was first anticipated. The design team was concerned about using low-e windows with low solar heat gain coefficients (SHGCs) on the south side of the homes. Working with three approaches—Energy 10 software, REMRate software, and “Passive-Solar Design Strategies,” published by the Sustainable Building Industries Council—they found little difference in annual energy bills between windows with similar U-values and different SHGCs.

In general, the savings in the winter provided by the higher SHGC were balanced by higher cooling bills in the summer. Cooler climates, such as Asheville, showed somewhat higher performance for the higher-SHGC windows, while in warmer climates, such as Wilmington, the lower-SHGC windows performed better. However, the differences in performance were no greater than 2% of total heating and cooling bills. Therefore, we decided in most cases to go with standard low-e, which has an SHGC of 0.40 or less. In the Oaks, the low SHGC will help provide comfort during summer months since the homes do not have air conditioning systems.

Solar Water Heating

ASU research staff bid out the solar water-heating systems for the first two homes. The bids for installed systems came in higher than expected, averaging \$4,500 each. In order to reduce the cost of water heating and provide a learning opportunity to students in the Department of Technology, the project staff decided to order the parts and install the systems themselves. The cost for each system, uninstalled, was \$2,500. The installation, by the ASU team members and student volunteers, went quite well, and the systems are now cranking out the Btu.

The economics of solar water heating are easier to justify in North Carolina than in most other states, because the state



ROB HOWARD

Energy efficiency features in the Zero-Energy Home include 2 x 6 walls with Icynene foam insulation.



ROB HOWARD

Catawba Valley Habitat for Humanity became an enthusiastic partner in the affordable passive solar home project.

has maintained a solar tax credit, now 35% of the initial cost up to a maximum credit of \$1,400. Assuming an increase in the demand for solar water heaters, if the installed system cost can be reduced to \$3,500, the state solar tax credit removes \$1,225 from this cost, for a net system cost of \$2,275. The annual energy savings from the solar water-heating systems for homes with electric water heating will be \$240, with a payback period of nine years. One of the goals of projects such as this is to increase the market penetration of solar water heating. Both ASU and the North Carolina Solar Energy Center are researching options for reducing the cost of solar water heating systems.

The recently passed National Energy Bill will further improve the economics

of solar water heating. The bill provides a 30% tax credit. Thus the \$3,500 system described above will benefit from a total credit in North Carolina of 65%, or \$2,275, resulting in a net installed cost of only \$1,225 and a payback period of about five years.

The bill also provides a substantial incentive for homes that reduce heating and cooling costs 50% below those for similar homes that exactly meet the 2003 International Energy Conservation Code (IECC). All of the homes built according to the specifications of the project should qualify for this incentive—currently pegged at \$2,000 per home.

Working Out the Kinks

The initial two homes at The Oaks posed their share of challenges, as the builder and the design team learned together. One of the important lessons learned was that energy details shown on the plans and taught in the classroom are not always carried out in the field. For example, the builder insulated the first two feet horizontally under the slab, but left out the critical vertical perimeter insulation on the first home. The situation was remedied, but not without spending extra time and materials. In

addition, the ceiling framing plan provided pathways to keep the ductwork inside the building envelope. However, the plan was not fully followed, which meant that the ductwork layout had to be redesigned in a more complex arrangement. The team also discovered that the solar water heater's pressure relief valve leaked whenever the water turned on, because local water pressure is high and the plumbing system did not initially include an expansion tank.

The first home that the team worked on was to have ceramic tile for the floor finish over the concrete slab. However, during the construction coordination process, it was changed to stick-on laminated wood tile. Then the laminated tile buckled, so the builder replaced it with the ceramic tile as originally designed.

The first home cost more than expected, which raised some eyebrows at the housing agency. The original design for the site, a non-passive home with virtually the same window area as the passive solar design, was a single-story ranch-style home. While some attributed the added cost to the passive solar and green building features, most recognized it was the more stylish two-story design along with the fact that a number of months had passed since the initial bid.

Catawba Valley Habitat for Humanity

As mentioned earlier, ASU's contract with the state energy office included funding for the renewable-electricity features of a Net Zero-Energy Home. Catawba Valley HFH readily agreed to build the home, recognizing that the design and construction process would most likely present quirks and obstacles. The chapter recently won an Energy-Value Housing Award from DOE, in part due to the commitment of construction manager Rob Howard. Howard was the ultimate positive force on the work site; he kept the entire project moving on schedule while overcoming a host of unanticipated problems without sacrificing the high-performance features.

The features of the home ultimately qualified it for a Gold certification in

the North Carolina HealthyBuilt Homes program, as well as Energy Star and SystemVision approval (see “North Carolina Healthy Homes”). The energy efficiency and solar-thermal features of the Zero-Energy Home include

- 2 x 6 walls with Icynene foam insulation;
- insulated attic with R-50 blown-in fiberglass insulation;
- low-e double hung windows with a SHGC of 0.55 and a U-value of 0.34 donated by Andersen from their 400 series;
- simplified duct design with most ducts located inside the conditioned space;
- 1-ton geothermal heat pump for space heating and cooling;

• Energy Star refrigerator (Whirlpool ET8WTEMO—412 kWh/year) and dishwasher (Whirlpool DU915PWP—365 kWh/year);

• ultra-efficient, ventless clothes washer and dryer combination with estimated annual costs of \$9 with gas water heating and \$21 with electric water heating (Thor Laundry System model WD9900A);

• compact fluorescent lighting;

• enthalpy recovery ventilation system, which exchanges the heat and humidity in the ventilation exhaust airstream with that in the incoming air stream; and

• passive-solar design with concrete slab thermal mass—this time covered with ceramic tile made from recycled materials.

In order for a house to qualify as a Zero-Energy Home, its annual energy usage, although quite small, must be balanced by renewable-electricity production on the premises. ASU’s team designed and installed a 4.5 kW PV system with equipment for direct interconnection to the electric utility grid. Once again, a team of solar installers, local subcontractors, and ASU students went to work and installed the PV system, along with the solar water heating unit. The electrical and plumbing subcontractors oversaw the installation and made the final connections.

PV systems are typically difficult to justify economically, but the Catawba Valley Zero-Energy Home can potentially show a small positive economic return. Total cost of the PV system is estimated to be \$25,000. The actual cost was lower,



BRENT SUMMERVILLE

The Zero-Energy Home features a 4.5 kW PV system that offsets the home’s annual energy use.

Table 1. PV System Economic Analysis

	Cost (\$)
Equipment and installation cost	25,000
State solar tax credit	-8,750
Net photovoltaic system cost	16,250
Annual cost (net cost financed at 6% for 30 years)	1,169
Annual Electricity Generation Revenues	
Estimated annual generation (kWh)	6,652
Utility avoided cost revenues (\$0.035 per kWh)	233
NC GreenPower revenues (\$0.180 per kWh)	1,197
Total annual revenues	1,430
Net annual benefit	261

but it relied on donated labor as well as on labor costs covered by Energy Office funding. The project benefits from the NC GreenPower program, administered by Advanced Energy Corporation with participation from North Carolina’s electric utilities. NC GreenPower offers \$0.18/kWh for renewable electricity coming from PV systems. Cursory economic analysis showed that the home could generate about \$1,430 in electrical revenues annually from NC GreenPower and utility avoided cost payments (see Table 1). The annual additional mortgage and insurance costs for the system, after subtracting the state solar tax credit, would be about \$1,300, resulting in a slightly positive net cash flow. Obviously, without the state solar tax credit and the available green power rate, the economics would take a decided turn for the worse. Economics will be further helped by incentives in the National Energy Bill.

More Kinks

The construction of this home delivered the expected kinks and quirks, along with some painful and, at times, embarrassing lessons. The ASU team specified low-e windows on the southern exposure with relatively high SHGC of 0.55 to try to maximize heat gain in the winter. Most low-e windows have a SHGC of 0.4

or less. The window units were extremely difficult to find and order,

North Carolina Healthy Homes

The North Carolina HealthyBuilt Homes program, developed and operated by the North Carolina Solar Energy Center, provides a certificate for homes that meet green home guidelines, built by residential builders who practice sustainable, high performance building strategies.

To view the point system for the program, you can download the checklist, in Excel format, at http://healthybuilthomes.org/builders_and_designers.cfm. Just click on the Checklist on the right of the page.

even with the assistance of the window manufacturer. When the windows arrived, the SHGC was 0.38. The window company assured us that sashes containing windows with higher SHGCs would be delivered and could be swapped. They have yet to arrive.

The building inspector would not allow windows as close to the corner as the plans indicated. This would have eliminated one of the south-facing windows, even though the entire home was sheathed with 1/2-inch extruded polystyrene installed on top of 1/2-inch oriented strand board (OSB). The final compromise required installing 1/2-inch OSB on the interior of the south-facing wall.

The truss company delivered the wrong trusses. They were not as long as the plans specified and therefore did not provide as much overhang as designed. By the time the design team discovered the problem, the trusses were installed. The Habitat volunteer framing crew framed out an additional overhang to provide summer shading from direct sunlight.

The HVAC drawings indicated a very simple duct design, with most of the ductwork located inside the conditioned space. The HVAC contractor instead installed most of the ductwork—the supply trunk duct and all of the branch ducts—in the attic. Given these difficulties, the design team wished it could have afforded to spend every day on the job site. However, they found the Habitat crew more than willing to correct most of the problems.

Long-term maintenance is a key concern for any project involving non-standard technologies. In the case of The Oaks and Ridgeview Village, the management team at both projects has nearby offices. Officials with both groups have assured us that they will continue to monitor the project and make sure that any maintenance issues are resolved quickly.

The Road Ahead

Project staff at ASU are continuing their outreach and design efforts, working with affordable housing groups through-

out the region. The team hopes to have their designs in six to ten counties in the next year. In addition, the staff will collect and analyze energy bills and submetered energy use data using data logging equipment for the homes. They also intend to add more passive-solar home plans to their on-line set of plans.



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For more information:

Visit www.ncenergystar.org.

Check out ASU's on-line set of passive-solar home plans at www.energystar.org. Look under the "Solar Plan Book" tab.